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BTO-1 Lab Pack (Foundations)

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BTO-1 Lab Pack (Foundations) – Documentation

Title: Blue Team Lab Setup, Sysmon Integration, and Microsoft Sentinel Analytics Validation

Objective

To build foundational Blue Team skills by deploying and validating Microsoft Sentinel analytics, signals, and visualizations using a simulated enterprise data environment. This includes configuring Sysmon, Data Collection Rules (DCRs), and securely transferring Sysmon logs to the Sentinel workspace.

1. Signal Onboarding & Health

Goal

Enable Microsoft Sentinel and verify that telemetry is arriving from connected sources.

Steps

1. Open Microsoft Sentinel and connect it to your Log Analytics Workspace.
2. In Data Connectors, enable:
 - Azure Monitor Agent (AMA)
 - Security Events (Windows Event Logs)
3. Run the queries below to confirm heartbeat and security events are fresh:

Heartbeat

| summarize LastHeartbeat = max(TimeGenerated) by Computer

SecurityEvent

| summarize LastSecurityEvent = max(TimeGenerated) by Computer

The screenshot shows the Microsoft Sentinel interface with a query editor and results table. The query editor has a 'Run' button, a 'Time range' dropdown set to 'Last 7 days', and a 'Show' dropdown set to '1000 results'. The query text is as follows:

```
1 Heartbeat
2 | summarize LastHeartbeat = max(TimeGenerated) by Computer
3
4 SecurityEvent
5 | summarize LastSecurityEvent = max(TimeGenerated) by Computer
6
```

Below the query editor, there are tabs for 'Results' and 'Chart', and a link to 'Add bookmark'. The 'Results' tab is active, showing a table with the following data:

☐	Computer	LastSecurityEvent [UTC] ↑↓
☐	> naksmannBlueVm	2025/10/13, 13:48:18.225

4. Created a Health Workbook showing the freshness of these two sources.

Verification

- Each connected machine appears with a recent heartbeat timestamp.
- SecurityEvent logs are received within the past few minutes.

2. Windows Security via AMA

Goal

Validate collection of native Windows Security logs (Event IDs 4624, 4625, 4688).

Steps

1. Create a Data Collection Rule (DCR) for your Windows hosts.
 - Log Category: SecurityEvents
 - Destination: Your Sentinel workspace
2. Verify EventIDs 4624 (Logon), 4625 (Failed logon), and 4688 (Process Creation):

SecurityEvent

| where EventID in (4624, 4625, 4688)

| summarize count() by EventID

▶ Run

Time range : Last 7 days

Show : 1000 results

```
1 SecurityEvent
2 | where EventID in (4624, 4625, 4688)
3 | summarize count() by EventID
4
```

...

Results

Chart

🔖 Add bookmark

☐	EventID	count_
☐	> 4624	447
☐	> 4625	25238
☐	> 4688	66

Verification

- Count for each event ID > 0 confirms data is flowing successfully.

3. Sysmon + DCR (Sysmon Log Transfer Integration)

Goal

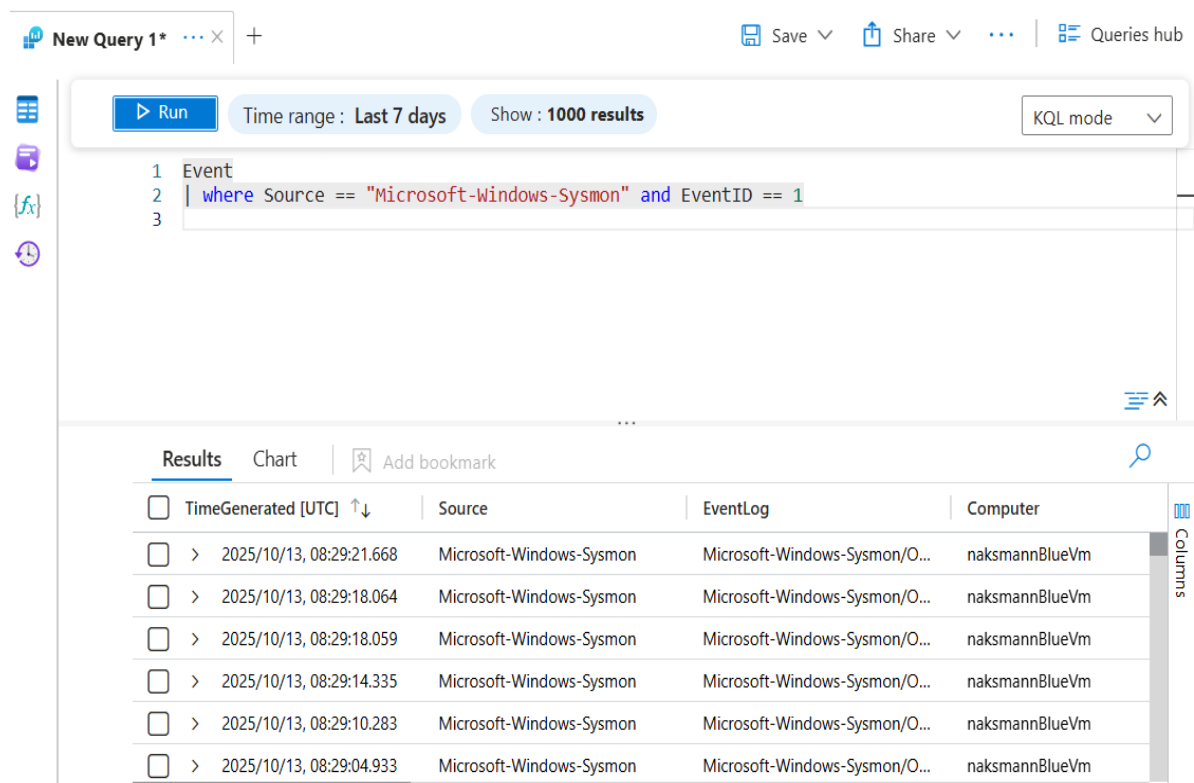
Enhance telemetry with Sysmon for detailed process and network monitoring, and successfully send Sysmon logs to the DCR you created for Microsoft Sentinel.

Steps

1. Installed Sysmon on my lab endpoint using the following command:
2. `sysmon.exe -i sysmonconfig.xml`
3. Created a new DCR to collect Sysmon logs:
 - Source: Microsoft-Windows-Sysmon/Operational
 - Destination: Your Sentinel workspace
4. Linked my endpoint to this DCR to ensure all Sysmon logs are being forwarded.
5. Verified Sysmon log transfer by running the query below:

Event

| where Source == "Microsoft-Windows-Sysmon" and EventID == 1



The screenshot shows the Microsoft Sentinel KQL query interface. At the top, there's a header with 'New Query 1*' and a plus icon. Below this, there's a toolbar with 'Run', 'Time range: Last 7 days', 'Show: 1000 results', and 'KQL mode'. The query editor contains the following KQL query:

```
1 Event
2 | where Source == "Microsoft-Windows-Sysmon" and EventID == 1
3
```

Below the query editor, there's a 'Results' tab with a search icon and 'Add bookmark'. The results are displayed in a table with the following columns: TimeGenerated [UTC], Source, EventLog, and Computer. The table contains six rows of data, all from the 'Microsoft-Windows-Sysmon' source on the 'naksmannBlueVm' computer.

TimeGenerated [UTC]	Source	EventLog	Computer
> 2025/10/13, 08:29:21.668	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/O...	naksmannBlueVm
> 2025/10/13, 08:29:18.064	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/O...	naksmannBlueVm
> 2025/10/13, 08:29:18.059	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/O...	naksmannBlueVm
> 2025/10/13, 08:29:14.335	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/O...	naksmannBlueVm
> 2025/10/13, 08:29:10.283	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/O...	naksmannBlueVm
> 2025/10/13, 08:29:04.933	Microsoft-Windows-Sysmon	Microsoft-Windows-Sysmon/O...	naksmannBlueVm

Verification

- Results show process image paths such as `C:\Windows\System32\notepad.exe`
- Confirms Sysmon telemetry is being collected and successfully transferred via your DCR.

4. KQL Essentials

Goal

Learn to query, filter, and summarize data using KQL.

Starter Queries

a. Using where:

```
SecurityEvent  
| where EventID == 4625
```

b. Using summarize:

```
SecurityEvent  
| summarize FailedLogons = count() by Computer
```

c. Using bin:

```
SecurityEvent  
| where EventID == 4625  
| summarize count() by bin(TimeGenerated, 1h)
```

5. OOB Rule Enablement

Goal

Enable Out-of-the-Box (OOB) analytics rules to automatically detect threats.

Steps

1. Went to:

Sentinel
Analytics
Rule Templates.
2. Enabled the following example rules:
 - Brute Force Attack against Windows Local Accounts
 - Suspicious Process Execution (Sysmon)
3. Confirm their status = Enabled.

	 Med...	Brute-Force alerts	 S...	 Enabled		Custom Content	10/12/2025, 4:59...
	 Low	Excessive Windo...	 S...	 Enabled	 Credential Acci	T1110	Windows Securit... 10/10/2025, 9:35...
	 Med...	Process Executio...	 S...	 Enabled	 Execution	T1059	Windows Securit... 10/10/2025, 9:33...

Verification

- Alerts appear in Incidents after simulating failed logons or suspicious process activity.

6. Workbook 101

Goal

Built visual dashboards using KQL queries.

Steps

Went to:

Workbooks(+ Add new ,Blank Workbook).

Added these visualizations:

a. Failed vs Successful Logons

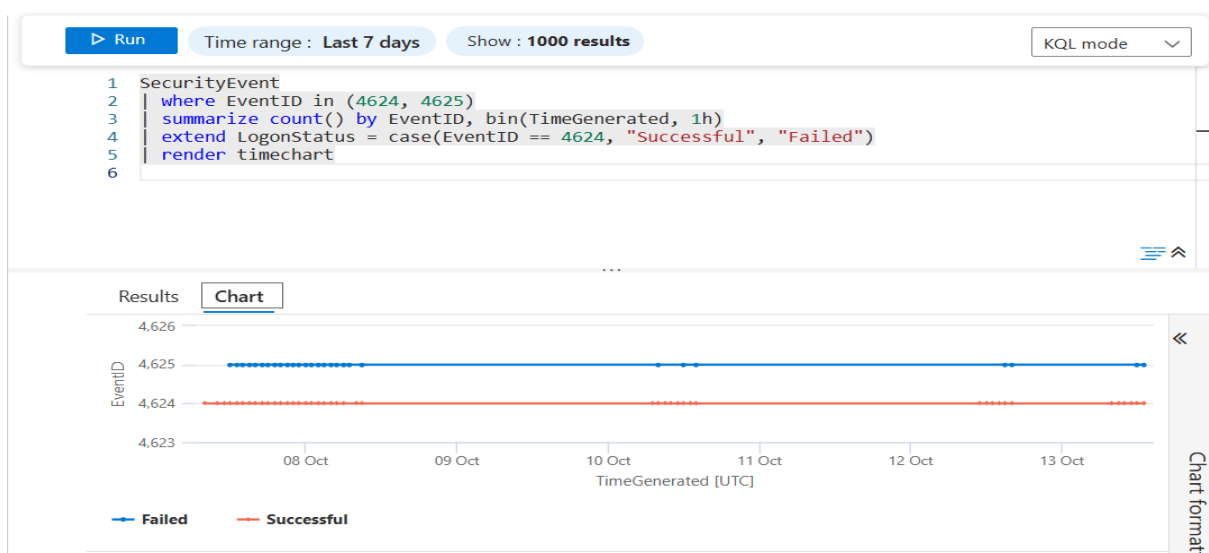
SecurityEvent

| where EventID in (4624, 4625)

| summarize count() by EventID, bin(TimeGenerated, 1h)

| extend LogonStatus = case(EventID == 4624, "Successful", "Failed")

| render timechart



b. Sysmon – Top N Processes

Event

| where Source == "Microsoft-Windows-Sysmon" and EventID == 1


```

| extend parsed = parse_xml(EventData)
| extend EventData = parsed.EventData.DataItem.EventData
| mv-expand Data = EventData.Data
| extend Name = tostring(Data['@Name']), Value = tostring(Data['#text'])
| where Name == "Image"
| summarize count() by Image = Value
| top 10 by count_
| render barchart

```

Verification

- Workbook shows a time chart for logons and a bar chart for top processes.

7. Watchlist 10

Goal

Import high-value user or asset lists and use them for monitoring and correlation.

Option A — High-Value Users

CSV Used:

```

SearchKey,AssetRole
naksmann_03,IT Administrator
naksmann_04,C-Level Executive
naksmann_05,Finance Manager

```

Steps

1. Upload CSV in Sentinel (Configuration, Watchlists, Add new watchlist.)
 - Name: HighValueUsers
2. Use the query below to correlate events:

```

let hv = _GetWatchlist('HighValueUsers');
SecurityEvent
| where TimeGenerated > ago(1d)
| where EventID in (4624, 4625, 4688)
| extend AccountShort = tostring(split(Account, '\\')[-1])
| lookup kind=leftouter hv on $left.AccountShort == $right.SearchKey
| summarize cnt = count() by AccountShort, AssetRole, bin(TimeGenerated, 1h)
| sort by cnt desc

```

Result: Displays logon and process events for high-value users.

Option B — High-Value Assets

CSV Used:

```

SearchKey,AssetRole
DC01,Domain Controller
FIN-SRV01,Finance Server
HR-SRV02,HR Server

```

Query

```

let hv = _GetWatchlist('HighValueAssets');
SecurityEvent
| where EventID in (4625, 4688)
| lookup kind=leftouter hv on $left.Computer == $right.SearchKey
| summarize cnt = count() by coalesce(Computer, 'unknown'), bin(TimeGenerated, 1h)

```

Result: Displays failed logons or process activity for critical servers.

Conclusion

This lab successfully demonstrated the deployment and integration of Microsoft Sentinel with various telemetry sources.

By setting up Sysmon and connecting it to the custom DCR, detailed endpoint activity (process creation, network events) was successfully captured and sent to Sentinel.